

CHEMISTRY BOARD QUESTION PAPER: JULY 2025**Time: 3 Hours Max. Marks: 70****GENERAL INSTRUCTIONS:**

1. The question paper is divided into four sections.
2. Section A: Q.1 (10 MCQs, 1 mark each) & Q.2 (8 VSA, 1 mark each).
3. Section B: Q.3–Q.14 (12 questions, 2 marks each; attempt any 8).
4. Section C: Q.15–Q.26 (12 questions, 3 marks each; attempt any 8).
5. Section D: Q.27–Q.31 (5 questions, 4 marks each; attempt any 3).
6. Log table allowed. Calculator not allowed.
7. Figures to the right indicate full marks.

SECTION – A**Q.1. Select and write the correct answer: (10×1=10)**

1. Coordination number of atoms in bcc lattice— (A)2 (B)4 (C)6 (D)8 (1)
2. Unit of Henry's Law constant— (A)mol L⁻¹ bar⁻¹ (B)mol⁻¹ L bar⁻¹ (C)mol L⁻¹ bar (D)mol⁻¹ L⁻¹ bar⁻¹ (1)
3. pH of human blood— (A)6.9 (B)7.0 (C)7.4 (D)8.1 (1)
4. Work done for isothermal expansion 20→25 dm³ at 1.5 bar— (A)-1.5 (B)-5 (C)-7 (D)-7.5 (1)
5. Electrolysis of aqueous NaCl— (A)H₂ at cathode (B)Cl₂ at cathode (C)H₂ at anode (D)O₂ at anode (1)
6. Rate constant unit = mol dm⁻³ s⁻¹ → order— (A)0 (B)1 (C)1.5 (D)2 (1)
7. General formula for interhalogens— (A)XX₂ (B)XX₃ (C)XX₄ (D)XX₆ (1)
8. Radioactive elements found in— (A)3d (B)4d (C)Actinoids (D)Lanthanoids (1)
9. Metal used in Grignard reagent— (A)Al (B)Mg (C)Na (D)Fe (1)
10. Dicarboxylic acid— (A)Acetic (B)Phthalic (C)Caproic (D)Citric (1)

Q.2. Answer the following: (8×1=8)

1. Write formula for percent atom economy. (1)
2. Name monomer of nylon-6. (1)
3. Number of chiral carbons in glucose. (1)
4. Complete: CH₃–C≡N + 4(H) → ? (1)
5. Reagent to convert phenol → benzene. (1)
6. IUPAC name of C₆H₄Cl–CO–CH₃. (1)
7. Oxidation state of Fe in [Fe(CN)₆]⁴⁻. (1)
8. Using 6 g O₂: 2ClF + O₂ → Cl₂O + OF₂, ΔH = +38.55 kJ — calculate ΔH° of reaction

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SECTION – B**(Any 8) (Marks: 16)**

- Q.3. Gold crystallises in fcc, edge 4×10^{-8} cm, $M=197$ g/mol. Calculate density. (2)
- Q.4. Define: (a) Isotonic solution (b) Colligative properties. (2)
- Q.5. $[H^+]=0.01$ M $\rightarrow$ Calculate pOH. (2)
- Q.6. Derive relation between ΔH and ΔU for gases. (2)
- Q.7. What is salt bridge? Write functions. (2)
- Q.8. Distinguish between order and molecularity. (2)
- Q.9. Explain anomalous behaviour of F: (i)Oxyacids (ii)Hydrides (iii)Electronegativity (iv)Electron gain enthalpy. (2)
- Q.10. Balance: (i) $K_2Cr_2O_7+KI+H_2SO_4 \rightarrow$ products (ii) $K_2Cr_2O_7+H_2SO_4+H_2S \rightarrow$ products. (2)
- Q.11. IUPAC name: (i) $[Cu(NH_3)_4]^{2+}$ (ii) $Na_3[AlF_6]$. (2)
- Q.12. Complete: (i) $CH_3-CH=CH_2+HBr \rightarrow ?$ (ii) $C_2H_5OH+SOCl_2 \rightarrow ?$. (2)
- Q.13. Preparation of phenol from isopropyl benzene (write equations). (2)
- Q.14. Write a note on haloform reaction. (2)

SECTION – C**(Any 8) (Marks: 24)**

- Q.15. Benzene–toluene solution (30% toluene). Vapour pressures: 118.2 & 36.7 mmHg. Calculate partial pressures. (3)
- Q.16. Define: (i)Buffer solution (ii)Solubility product (iii)Common ion effect. (3)
- Q.17. 32 g O_2 expands isothermally & reversibly 2 bar $\rightarrow$ 1 bar at 298 K. Calculate work. (3)
- Q.18. What is zero-order reaction? Derive integrated rate law. (3)
- Q.19. Structures: (a)Chlorous acid (b)Perchloric acid. (3)
- Q.20. (a)Uses of helium (b)Lanthanoid contraction. (3)
- Q.21. (a)Why d-block compounds are coloured? (b)Representation of sodium hexacyanoferrate(III) (c)Hybridisation & magnetic nature: $[Co(NH_3)_6]^{3+}$ & $[NiCl_4]^{2-}$. (3)
- Q.22. Convert: (a)Acetic acid $\rightarrow$ Ethyl acetate (b)Acetaldehyde $\rightarrow$ Acetaldoxime (c)Acetone $\rightarrow$ Propane. (3)
- Q.23. (a)Action of Hinsberg reagent on ethylamine (b)Action of HNO_2 (c)Order of basicity: $CH_3NH_2, (CH_3)_2NH, (CH_3)_3N$. (3)
- Q.24. (a)What are monosaccharides? (b)Zwitter ion of amino acid (c)Nucleotide components. (3)
- Q.25. (a)Preparation of Teflon (b)Monomer of natural rubber (c)Uses of HDP. (3)
- Q.26. Explain: (a)Sustainable development (b)Nanomaterial (c)Gamma isomer of BHC. (3)



SECTION – D**(Any 3) (Marks: 12)**

Q.27. (a)Crystalline vs amorphous solids (b)Bleaching action of chlorine. (4)

Q.28. (a)Diagram: Standard Hydrogen Electrode (b)Calculate EMF: $\text{Mg}|\text{Mg}^{2+}||\text{Ag}^+|\text{Ag}$, $E^\circ = -2.37 \text{ V} \text{ \& } +0.80 \text{ V}$. (4)

Q.29. (a)Spontaneity: $\Delta H = +50 \text{ kJ}$, $\Delta S = -130 \text{ J/K}$ at 250 K (b)Optical activity of 2-chlorobutane (c)Full form of DDT. (4)

Q.30. (a)Coordination number (b)Reimer–Tiemann reaction (c)Structure of methyl vinyl ether. (4)

Q.31. (a)Aldol condensation of ethanal (b)Properties of interstitial compounds. (4)

